

Air University



Organization & Assembly (Lab)

Year 2025

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Lab Task # 07

Write a program in EXE template that takes a lowercase letter from the user and converts it into an uppercase letter. Display the result on the screen. Use LEA command to get the message rather than MOV command. Use EQU directive to assign a constant name to 0DH and 0AH.

Code

```
.MODEL SMALL          ; Define small memory model for the program
.STACK 100H           ; Allocate 256 bytes for the stack

.DATA
MSG1 DB 'Enter a lowercase letter: $'
MSG2 DB 0DH,0AH, 'The uppercase letter is: $'
CR EQU 0DH            ; Define carriage return constant
LF EQU 0AH            ; Define line feed constant

.CODE
MAIN PROC
    MOV AX, @DATA      ; Load data segment address into AX
    MOV DS, AX         ; Set DS to point to data segment

    ; Display prompt message
    LEA DX, MSG1        ; Load effective address of MSG1 into DX
    MOV AH, 09H
    INT 21H

    ; Read character from keyboard
    MOV AH, 01H         ; Function to read character with echo
    INT 21H

    ; Store the lowercase letter temporarily
    MOV BL, AL          ; Move input character to BL for processing

    ; Convert lowercase to uppercase (only if in 'a'..'z')
    CMP BL, 'a'
    JB NOT_LOWER
    CMP BL, 'z'
    JA NOT_LOWER
    SUB BL, 20H          ; Subtract 32 to convert to uppercase

NOT_LOWER:              ; Label for non-lowercase characters
    ; Display result message
    LEA DX, MSG2        ; Load effective address of MSG2 into DX
    MOV AH, 09H
    INT 21H

    ; Display the uppercase character
    MOV DL, BL          ; Move converted character to DL
    MOV AH, 02H
    INT 21H
```

```
; Display newline
    MOV DL, CR      ; Load carriage return into DL
    MOV AH, 02H
    INT 21H
    MOV DL, LF      ; Load line feed into DL
    MOV AH, 02H
    INT 21H

; Exit program
    MOV AH, 4CH
    INT 21H
MAIN ENDP
END MAIN
```

Output

```
D:\>D:\test
Enter a lowercase letter: h
The uppercase letter is: H
```

THE END
